

EEET202

INES & TRANSFORMERS

MODULE:3

Torque Equation of DC Motor and

Back EMF

of operation of DC motors and select appropriate motor types for

Prepared By,

ADARSH SR

Asst. Professor

EEE Department

SCET Kodakara



Syllabus

generation of torque, torque equation, performance characteristics – starters – 3 point and 4 point starters (principle only).

tors - field control, armature control. Braking of dc motors. Power and efficiency. Testing of dc motors - Swinburne's test, Hopkinson's test. DC motor applications – numerical problems.

INTRODUCTION

Principle of DC Motor

A DC motor converts DC power into mechanical power is called a DC motor.

It is based on the principle that when a current carrying conductor is placed in a magnetic field, the conductor experiences a mechanical force.

If this force can be given by Fleming's left hand rule, then it is called a DC motor.

When by, $F = B I l \sin \theta$

Where,

B: Magnetic field strength

I : Current in armature conductor

l : Length of conductor in meter

Θ : Angle between B and I

BACK EMF

of DC motor rotates under the influence of the
the armature conductors move through the magnetic

aday's law of Electromagnetic induction, an EMF is
mature similar as in the case of generator.

opposes the cause producing it according to Lenz's

s the applied voltage and hence induced EMF acts
applied voltage(V).

n as **Back EMF or Counter EMF**.

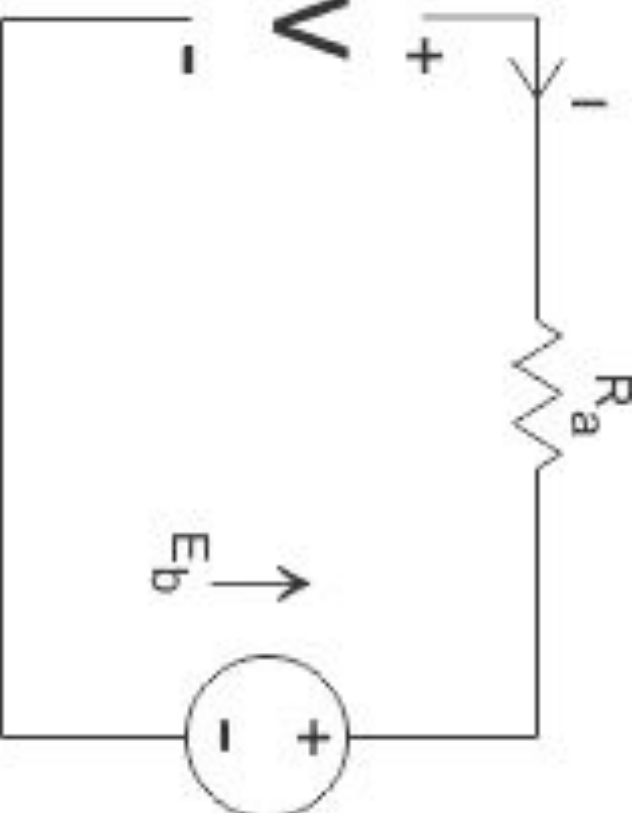
ck EMF is given by,

$$E_b = \frac{Z\phi N P}{60 A}$$

BACK EMF

$$= E_b + I_a R_a$$

can be given as, $I_a = \frac{V - E_b}{R_a}$



Torque Equation

umber of armature conductors

magnetic field strength in Wb/m^2

urrent in armature conductor (I_a/A)

lux per pole in Wb

umber of poles

umber of parallel paths

adius of armature core

length of armature conductor

Equation (contd..)

ing or twisting moment of a force about an axis.

ment of a force about an axis and is measured by the product of force (F) and to which the force acts i.e.

uctor is acted upon by a circumferential force F at a distance r , the radius of flow.

Use Equation (contd..)

on a single armature conductor, $F = BI l \sin \theta = BI l$
 ch conductor, $T = BI l \times R$

d on armature, $T = BI l R \times Z$

$$T = Z \frac{\phi}{a} \frac{I_a}{lR}$$

cross-sectional area of flux path per pole and

$$a = \frac{2\pi Rl}{P}$$

developed, $T_e = Z \frac{\phi}{2\pi Rl} \frac{I_a}{lR} = ZP \frac{\phi}{2\pi} \frac{I_a}{A}$

$$\left(\frac{2\pi Rl}{P} \right)$$

$$T_e = \frac{Z\phi I_a}{2\pi} \frac{P}{A} = 0.159 Z\phi I_a \frac{P}{A}$$

$$\phi I_a \frac{P}{A}$$

otor

radius of armature in m

length of each conductor in m

ber of armature conductors

f parallel paths

each conductor = I_a/A

ux density in Wb/m²

ple in Wb

f poles

conductor, $(F) = B i l$ newtons

one conductor = $F \times r$ newton- meter

torque, $(T_a) = Z F r$ newton- meter

$i l r$ newton- meter ... (1)



each conductor)

the x-sectional area of flux path per pole a radius r .

$$a = \frac{2\pi Rl}{P}$$

a cross-sectional area of flux path per pole and

in equation (1),

$$\frac{1}{r} \times l \times r$$

$$\times \frac{I_a}{A} \times l \times r$$

$$\times \left(\frac{P}{A} \right)$$

s(P) and number of the parallel paths (A) is constant,